

OFFICIAL TRAINING MANUAL & SYSTEM COURSE

MEGALODON

Self-Hosted API Security, Real-Time Reverse Proxy Gateway & Complete Host Network Visibility Platform

TARGET AUDIENCE

SecOps, DevOps & SREs

PLATFORM VERSION

Megalodon v1.0 Production

DESIGN AESTHETIC

OpenAI Monochrome
Minimalist

SYSTEM MISSION & SCOPE

Megalodon is an enterprise-grade, high-performance security platform and host observability platform. It combines raw Linux socket-level interface telemetry, real-time WebSocket ingress streaming, a declarative boolean rule engine, CIDR-based firewall controls, automated threat workers, and cryptographic audit trails into a single unified control plane.

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Platform Architecture & Data Pipeline

Megalodon is built on a decoupled, asynchronous micro-service architecture capable of inspecting millions of requests with sub-millisecond overhead. The platform combines kernel-level network collection with user-space application firewalling.

- **Megalodon Core Engine & Traffic Inspector (:8000)**
High-performance async ASGI reverse proxy intercepting incoming HTTP traffic. Evaluates IP CIDR policies, visual security rules, and rate limits in-memory before upstream forwarding.

- **Host Network Agent (Daemon)**
Lightweight Python collector interfacing with Linux `/proc/net`, AF_INET netlink sockets, and ethtool to gather interface states, open sockets, and live TCP connections.

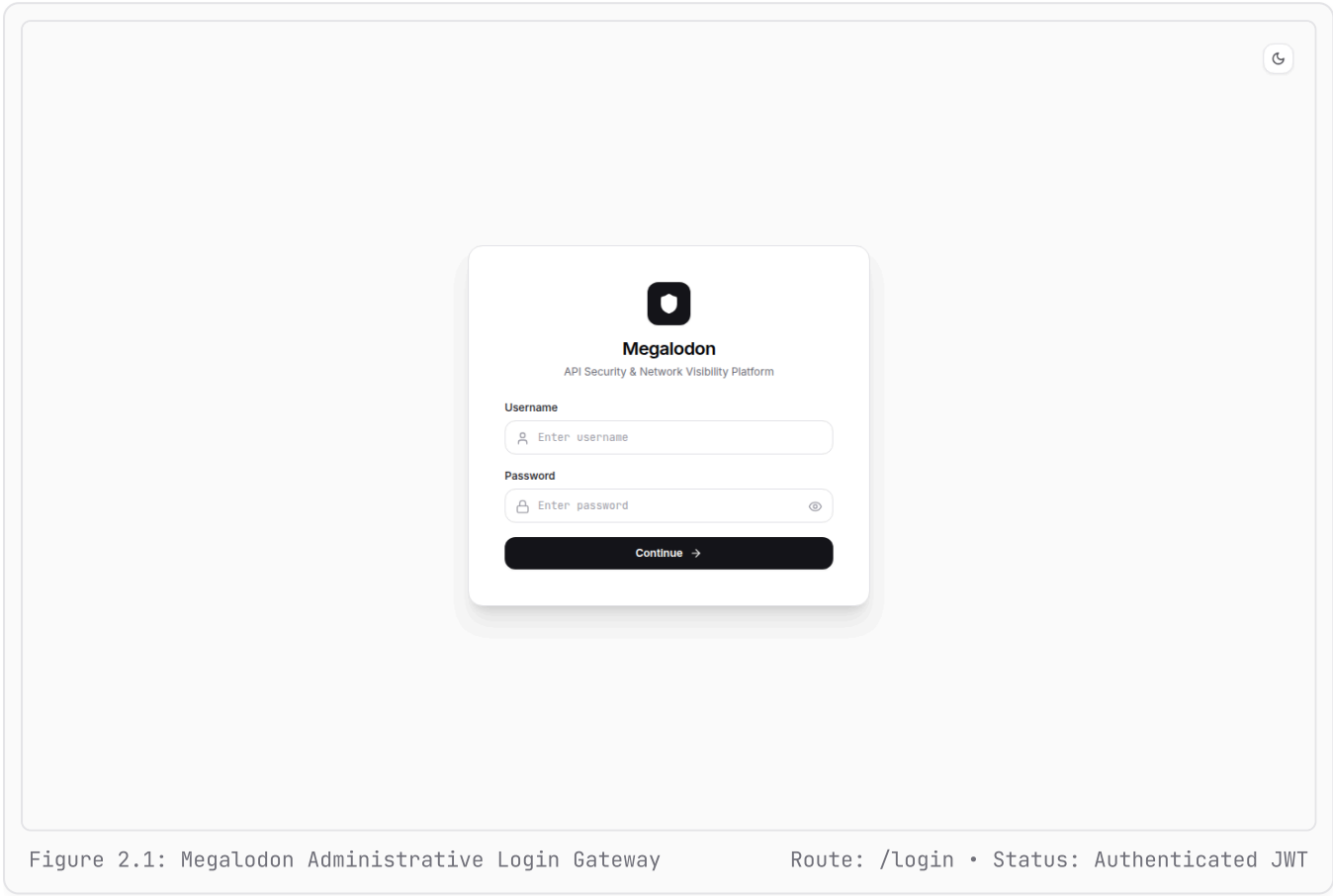
- **Event Streaming Pipeline (Kafka & Redis)**
Kafka buffers request ingress logs and security events with persistent partitions. Redis provides sub-millisecond caching for dynamic IP bans and rate-limiting buckets.

- **Dedicated Async Workers**
Specialized background workers consume Kafka topics: **Analytics Worker** (percentile calculations), **Security Worker** (attack signature heuristics), and **Alert Worker** (incident notification).

SERVICE COMPONENT	INTERNAL PORT	EXTERNAL PORT	PRIMARY RESPONSIBILITY
megalodon-frontend	80/tcp (Nginx)	:3000	React 18 SPA, monochrome minimalist dashboard, and WebSocket live feed
megalodon-api	8000/tcp (FastAPI)	:8000	Core platform engine, traffic filtering, rate limiting, and security rules API
megalodon-network-agent	Host Netlink / Proc	Host Daemon	Kernel network scanner, listening sockets, active connections & PID mapping
megalodon-postgres	5432/tcp	:5432	PostgreSQL 16 storage for rules, users, and immutable audit logs
megalodon-redis	6379/tcp	:6379	In-memory distributed rate limiter (Token Bucket / Sliding Window) & IP cache
megalodon-kafka	9092/tcp	:9092	Apache Kafka event streaming bus (KRaft mode, no ZooKeeper required)
megalodon-analytics-worker	Background Daemon	Async Consumer	Asynchronous Kafka consumer computing sliding-window latency percentiles
megalodon-security-worker	Background Daemon	Async Consumer	Automated threat detection and malicious IP quarantine manager
megalodon-alert-worker	Background Daemon	Async Consumer	Dispatches security incident alerts and operational notifications
megalodon-demo-users	5001/tcp	:5001	Upstream sample microservice used for routing and policy validation

Authentication & Dual-Theme Login Page

The Megalodon login gateway controls administrative access to the platform. It features JSON Web Token (JWT) stateless authentication, brute-force mitigation, and a persistent client-side monochrome theme switcher.



• **Dual-Mode Monochrome Theme**

Features a top-right instant theme toggle supporting both Dark and Light aesthetics conforming to OpenAI's minimalist monochrome design system.

• **Token-Based Session Management**

Issues HMAC-SHA256 signed JWT tokens stored securely in localStorage. API clients auto-inject bearer credentials on every authenticated request.

• **Quick Demo Sign-In**

Includes a one-click demo sign-in credential injector for immediate lab onboarding and testing with the default administrator account (`admin`).

• **Automatic Session Revocation**

Listens to `auth:unauthorized` global window events. If a token expires or is rejected with 401, the user is immediately redirected to login.

Central Operations Dashboard

The Operations Dashboard provides real-time situational awareness over both host infrastructure and application gateway ingress. It continuously aggregates latency percentiles, error rates, and security enforcement metrics.

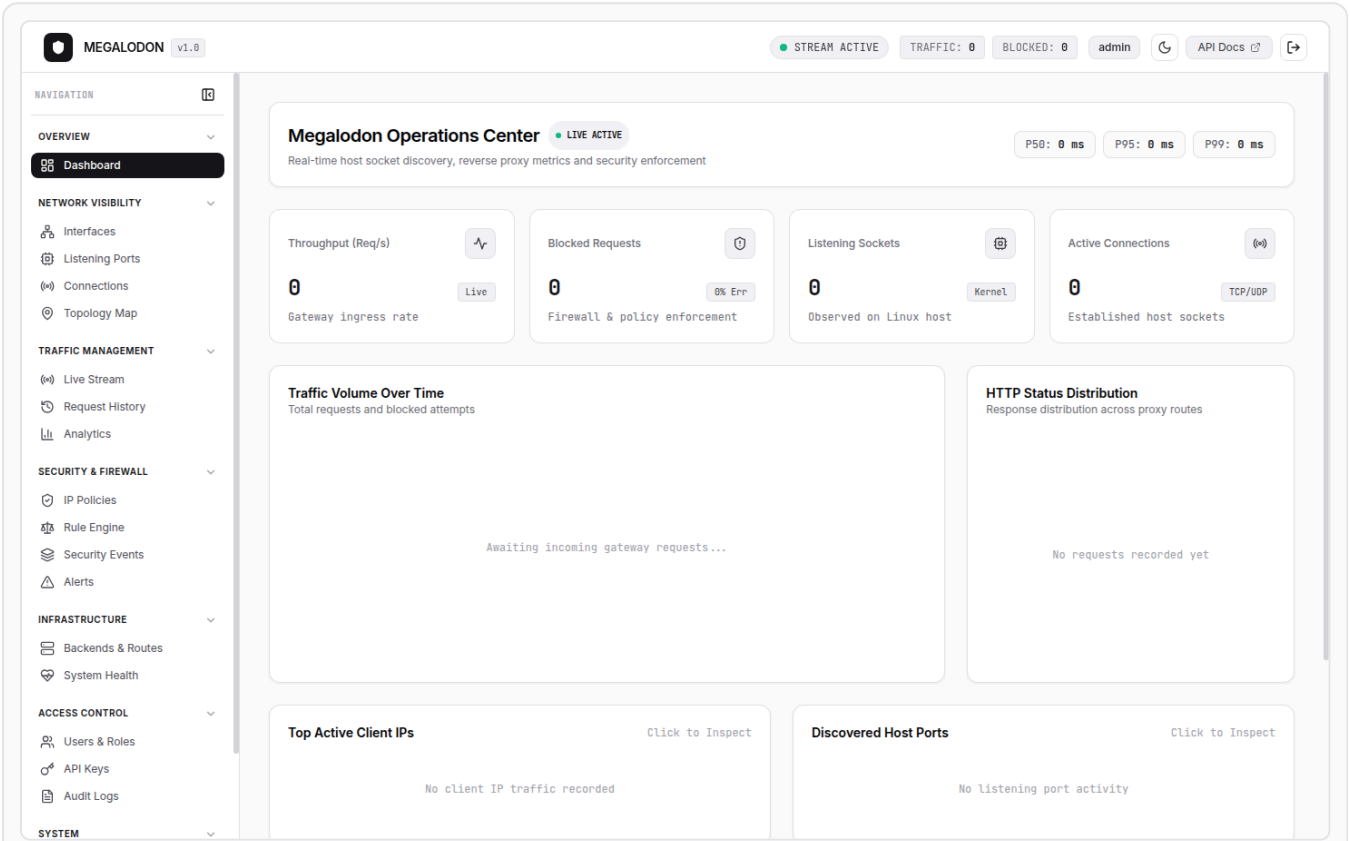


Figure 3.1: Central Operations Dashboard & Real-Time Telemetry Route: / • WebSocket Live Ingress

● **Executive KPI Metrics Banner**

Tracks Total Ingress Requests, Active Sockets, System Interfaces, and Security Blocks. Includes real-time status pulses and operational state indicators.

● **Latency Percentiles (P50, P95, P99)**

Interactive SVG AreaChart monitoring gateway round-trip time with real-time sliding time windows, highlighting latency anomalies.

● **Top Ingress Sockets & Client IPs**

Dynamic ranking of the top origin client IPs and active destination listening ports with request volume distributions.

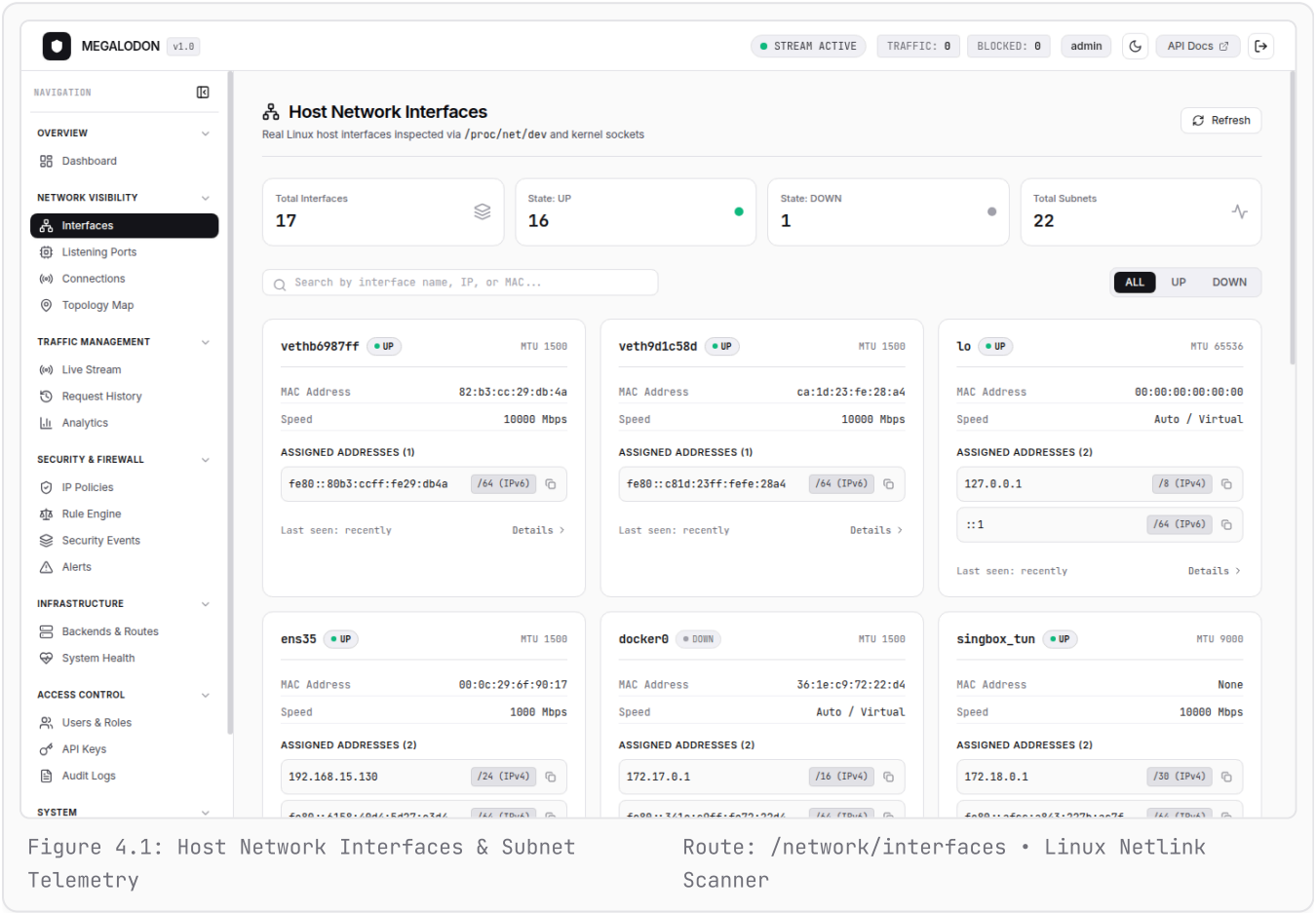
● **Live Ingress Stream Feed**

Embedded live micro-table displaying HTTP method, URI path, client source, latency (ms), and policy verdict (ALLOWED/BLOCKED) in real-time.

MODULE 04 • NETWORK VISIBILITY

Host Network Interfaces & Subnets

The Network Interfaces view audits all physical and virtual network adapters attached to the host operating system. It parses operational flags, MAC addresses, MTU payloads, and assigned IPv4/IPv6 CIDRs.

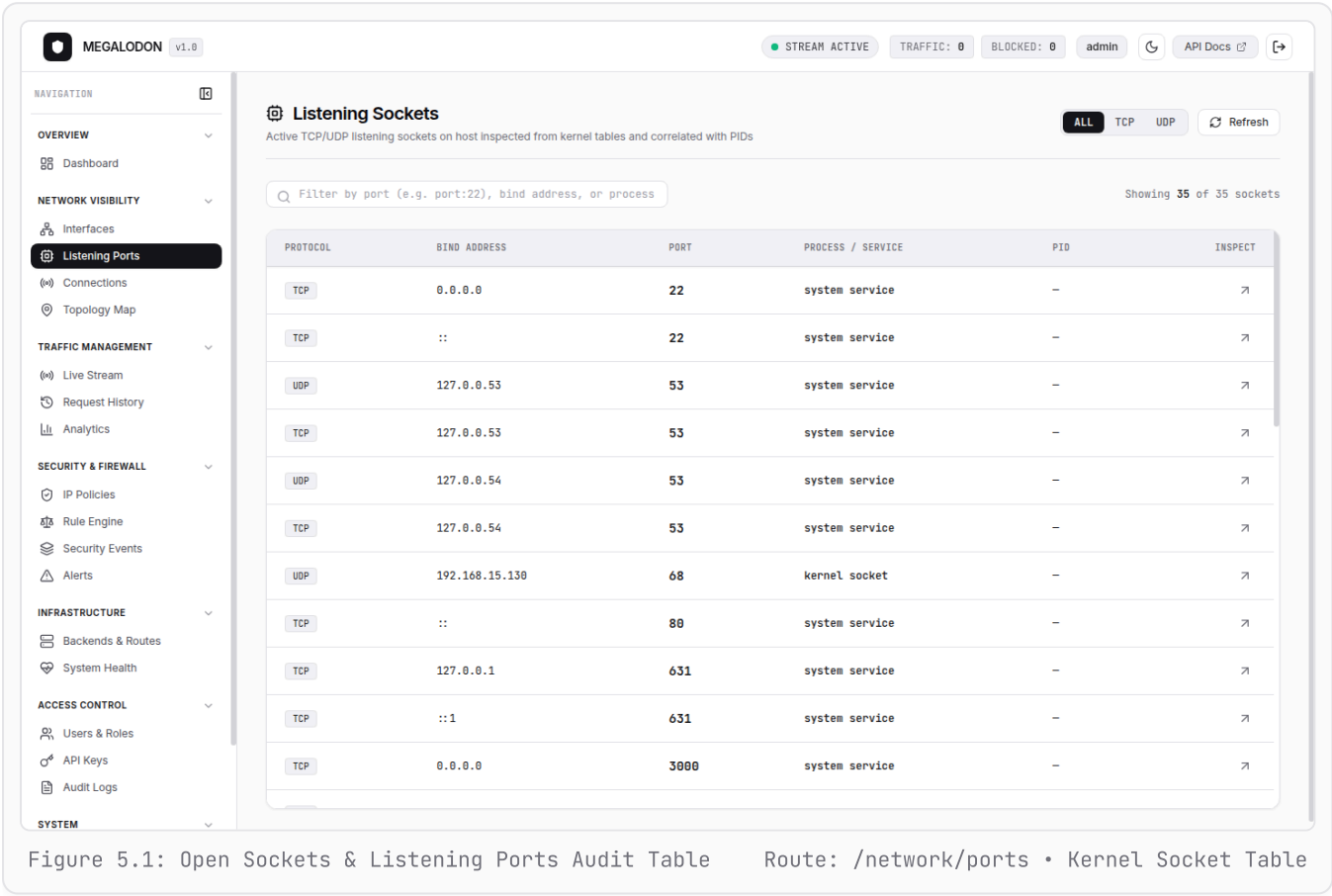


- **Interface Operational State**
Real-time detection of link states (`UP` / `DOWN`) across physical NICs (eth0, ens3), bridge devices (docker0, br-lan), and loopback interfaces.
- **Multi-IP CIDR Mapping**
Discovers all IPv4 and IPv6 addresses bound to each interface, including netmask prefix lengths and broadcast domains.
- **MTU & Link Speed Inspection**
Displays Maximum Transmission Unit (MTU) byte limits and interface negotiated speeds (Mbps) for throughput troubleshooting.
- **Interface Inspector Modal**
Detailed modal drawer portaled directly to `document.body` providing complete MAC address, driver properties, and address allocations.

MODULE 05 • NETWORK VISIBILITY

Host Listening Ports & Open Sockets

Auditing open network listening ports is essential to eliminating unauthorized attack surfaces. Megalodon scans all bound TCP/UDP listening endpoints and correlates them with host processes and active connections.



● Protocol & State Filtering

Filter across ALL, TCP, or UDP protocols. Instantly search by port number, bound IP address (e.g. 0.0.0.0, 127.0.0.1), or process name.

● Active Socket Connection Counter

Dynamically calculates how many active remote client sockets are currently connected to each open listening port.

● Port Detail Inspector Modal

Clicking any port opens an isolated modal listing all remote established TCP connections, peer IP addresses, ports, and send/receive byte queues.

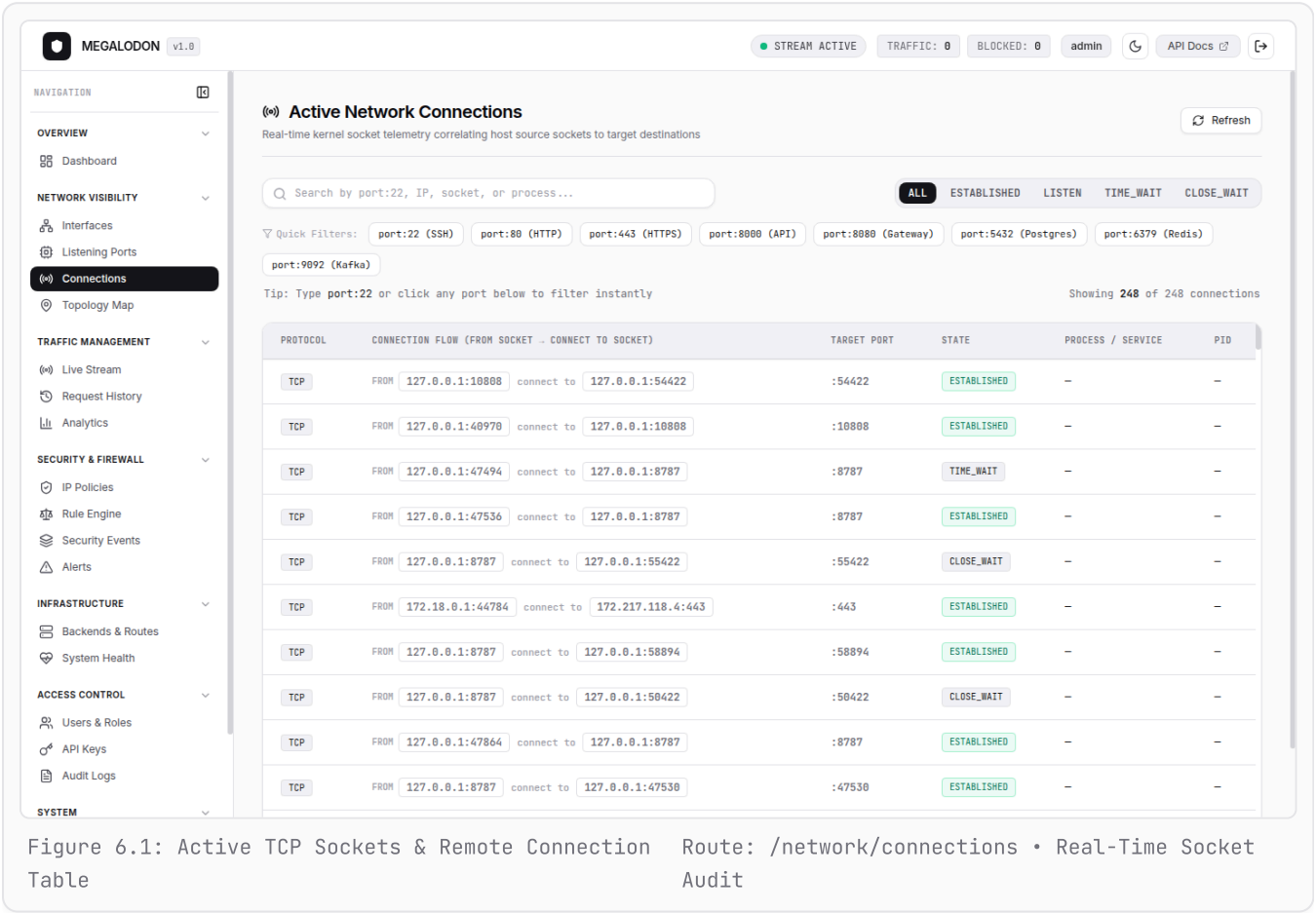
● Public Exposure Highlighting

Automatically flags sockets bound to 0.0.0.0 (wildcard listen) versus localhost-only listeners (127.0.0.1) for security verification.

MODULE 06 • NETWORK VISIBILITY

Active TCP Connections & Flow Tracker

Tracks active TCP state machines across the host (`ESTABLISHED` , `TIME_WAIT` , `CLOSE_WAIT` , `SYN_SENT`). Provides instant forensic inspection of client IP origins and suspicious outbound connections.



● Connection State Classification

Color-coded monochrome badges distinguishing live established flows from terminating or half-open connections.

● Remote IP Inspector Modal

Click any remote client IP to view aggregated request history, total bandwidth, active socket count, and one-click firewall ban capabilities.

● Port Search & Matching

Allows filtering by local port (e.g. searching `22` immediately isolates all active SSH sessions and origin client addresses).

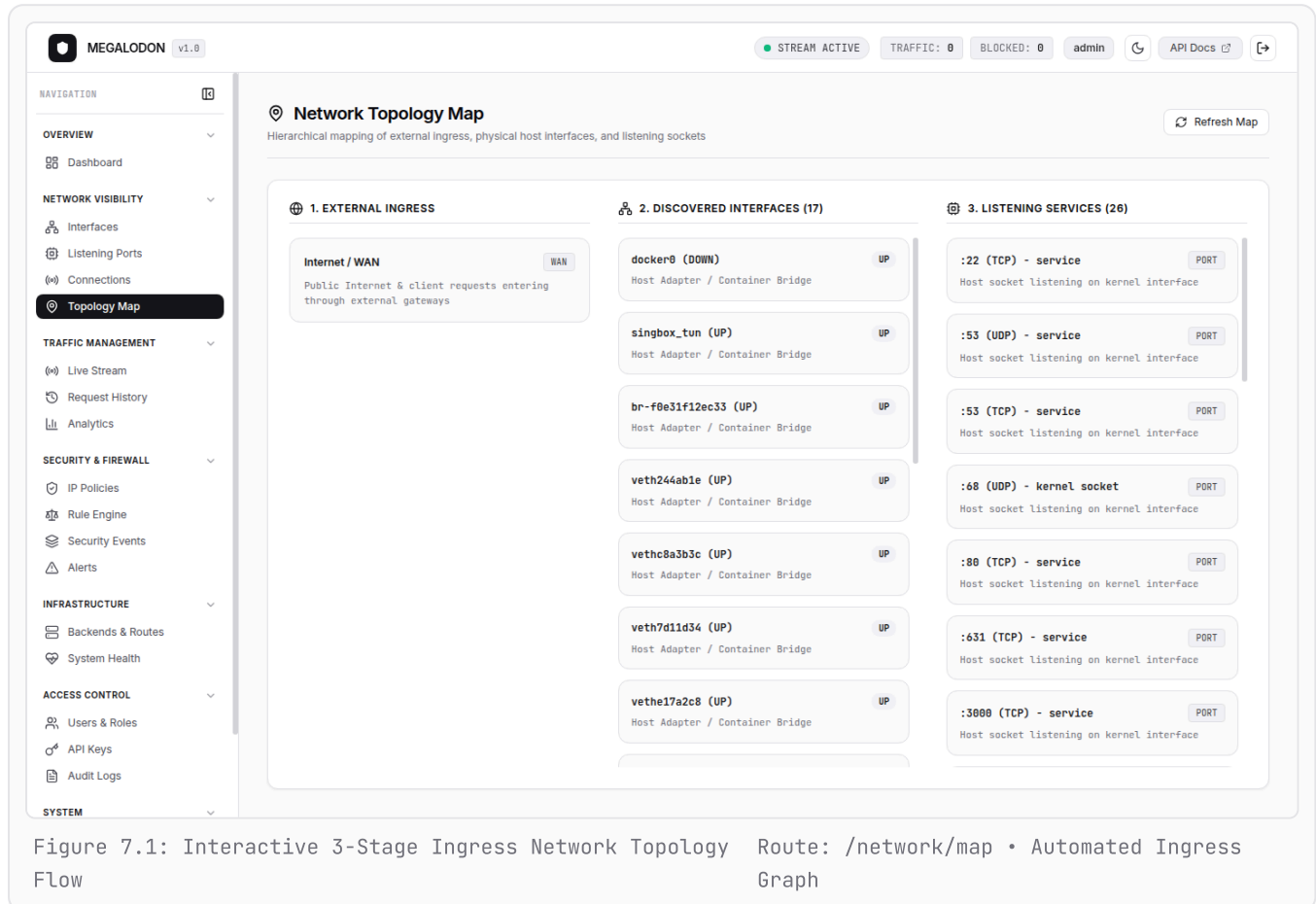
● Real-Time Flow Refresh

Continuous polling and on-demand refresh capability ensuring that transient connections are captured and audited immediately.

MODULE 07 • NETWORK VISIBILITY

Host & Ingress Network Topology Map

Visualizes the entire communication graph from external WAN ingress, through host network interface cards, and down into individual listening application sockets and backend containers.



● Stage 1: WAN Edge Ingress

Depicts external internet gateways, client traffic entry points, and incoming request volumes entering the host boundary.

● Stage 2: Discovered Host Interfaces

Visual representation of physical and bridge adapters with link states, packet delivery status, and assigned subnet addresses.

● Stage 3: Bound Application Sockets

Maps how interfaces route traffic directly to listening sockets (e.g. Megalodon Core Engine :8000, PostgreSQL :5432, Kafka :9092).

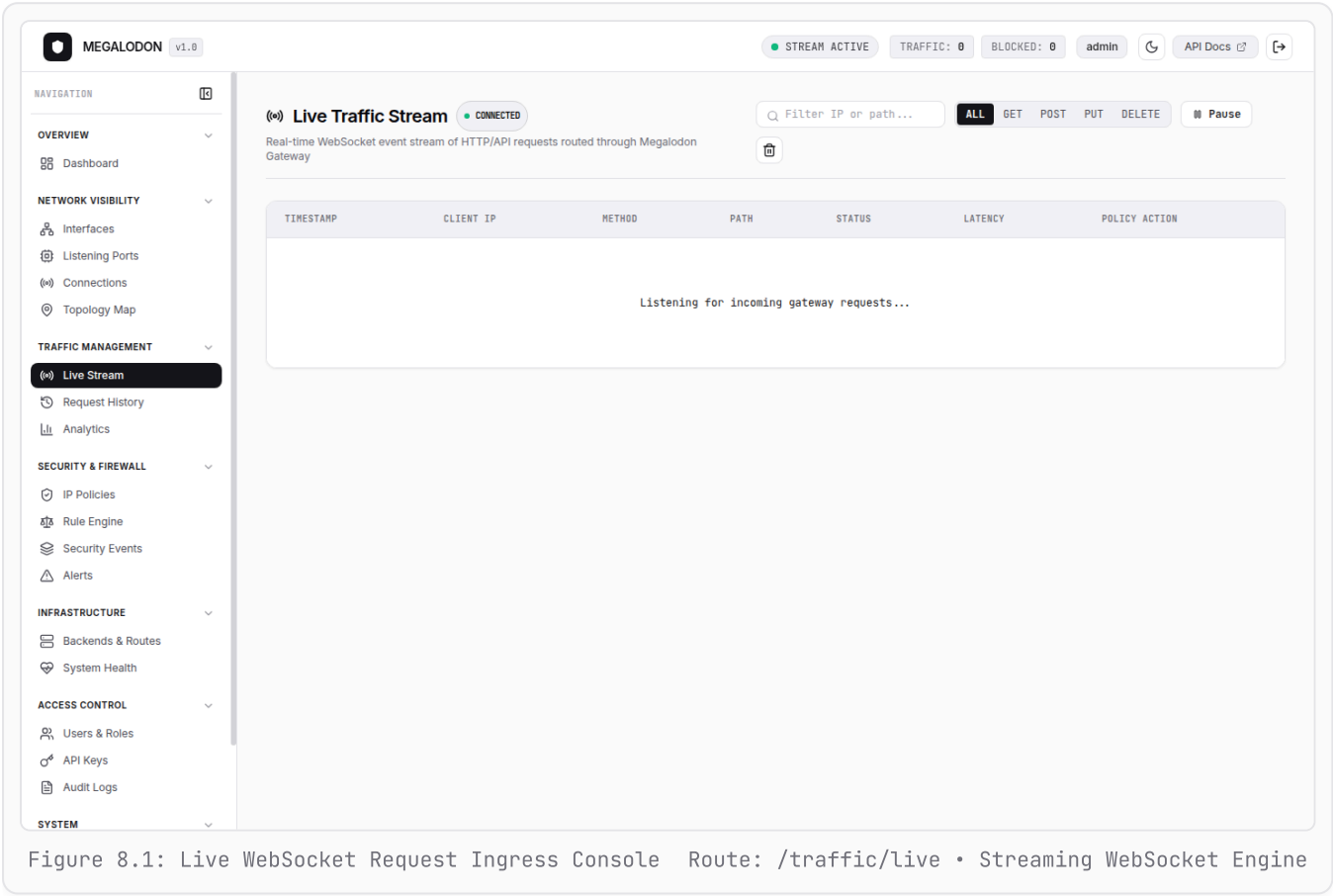
● Topology Graph Diagnostics

Provides node counters, link diagnostics, and real-time connectivity validation across the host's virtualized Docker networking fabric.

MODULE 08 • TRAFFIC MANAGEMENT

Real-Time WebSocket Traffic Stream

The Live Traffic Stream streams every single HTTP request passing through the Megalodon Ingress Engine in real-time over a persistent WebSocket connection (`ws://localhost:8000/api/v1/ws/traffic`).



● Stream Pause & Buffer Controls

Pause button allows operators to freeze the live display during heavy traffic surges to inspect individual request payloads without dropping connection.

● Immediate Policy Feedback

Each entry displays the precise enforcement verdict: `ALLOWED` (forwarded to upstream) or `BLOCKED` (mitigated at gateway).

● Live Search & Filtering

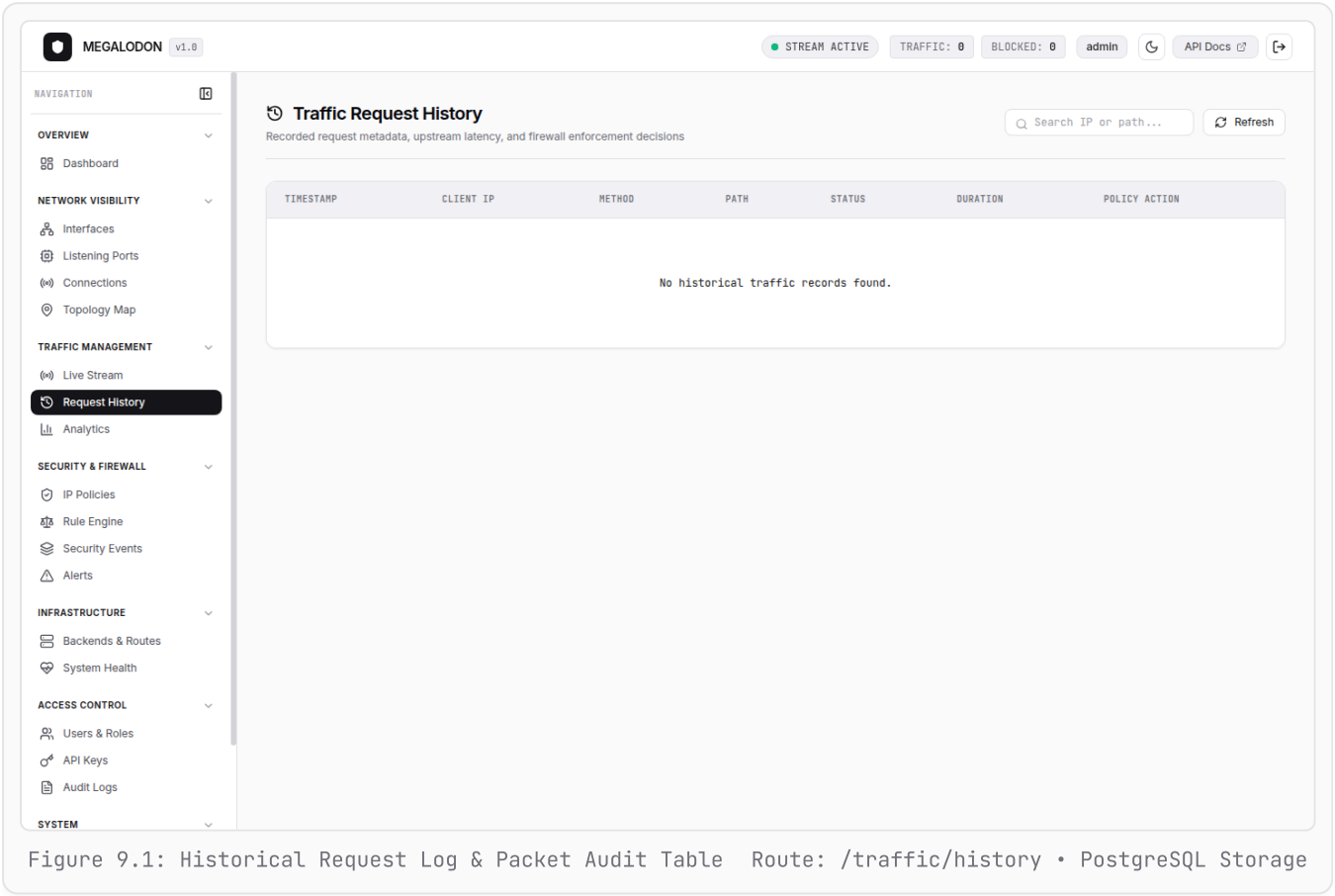
Instant real-time search across incoming paths, query parameters, HTTP methods (GET, POST, PUT, DELETE), and origin client IP addresses.

● Connection Heartbeat Monitoring

Header badge dynamically reports `STREAM ACTIVE` or `CONNECTING` with automatic exponential backoff reconnection.

Historical Request Ingress & Packet Logs

Provides deep forensic historical analysis of all requests processed by the platform. Stored durably in PostgreSQL with indexing on timestamps, client IPs, endpoints, and status codes.



● HTTP Status Code Auditing

Monochrome high-contrast badges for 2xx Success, 3xx Redirect, 4xx Client Errors (401, 403, 404, 429), and 5xx Upstream Server Errors.

● Sub-Millisecond Latency Profiling

Records exact execution time in milliseconds spent in the gateway reverse proxy versus upstream backend response time.

● Origin & Endpoint Filtering

Filter through historical logs by client IP, target host backend, HTTP method, or URI query pattern to reconstruct incident timelines.

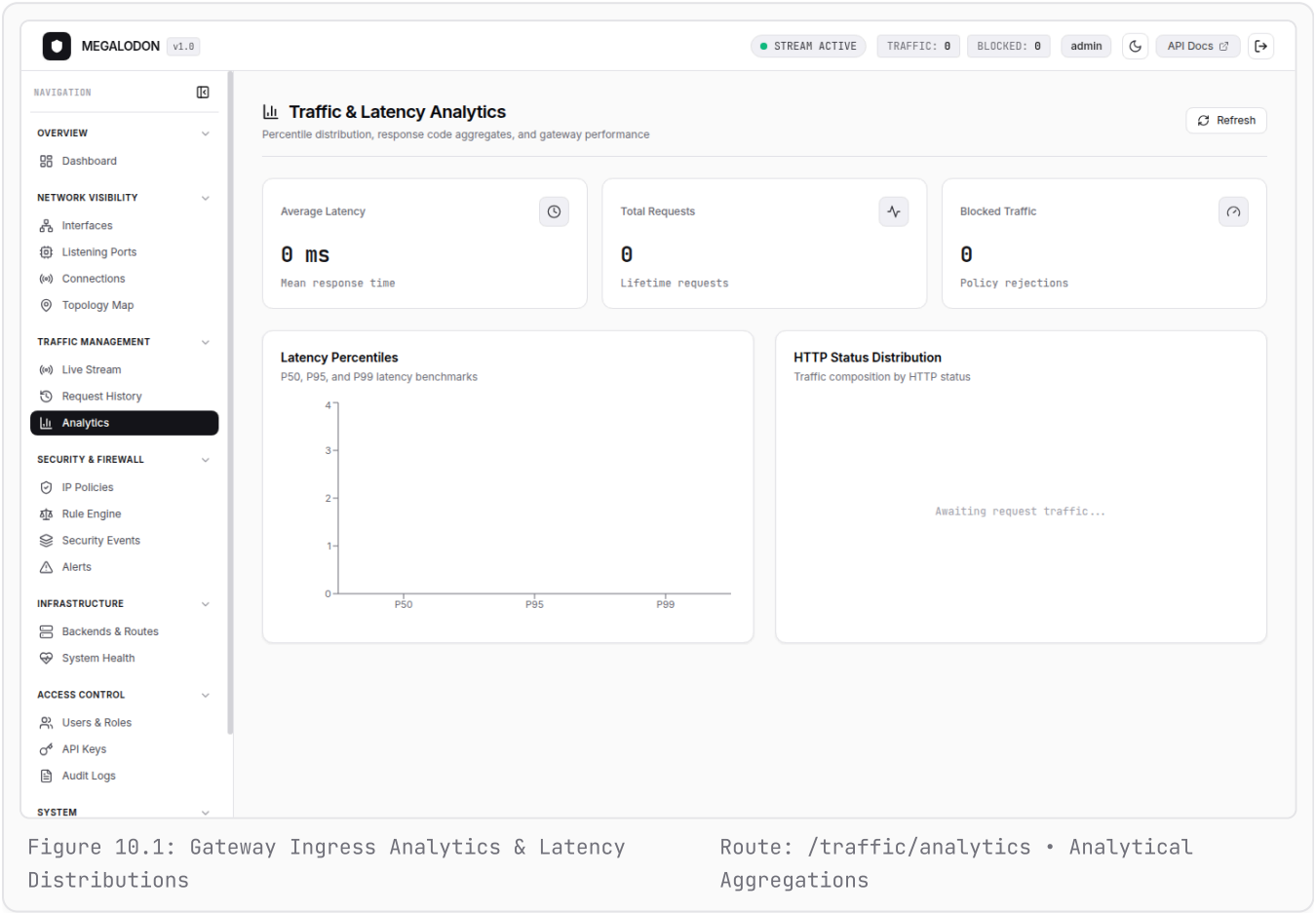
● Policy Enforcement Reason Tracking

Blocked requests detail the triggering policy rule ID, rate-limit threshold breach, or IP denylist reason for post-incident review.

MODULE 10 • TRAFFIC MANAGEMENT

Gateway Ingress & Latency Analytics

Transforms raw request streams into actionable statistical distributions. The Megalodon Analytics Service continuously processes sliding-window time buckets to compute percentiles and throughput trends.



● P50 / P95 / P99 Latency Curves

Visualizes median, 95th, and 99th percentile response times, allowing engineers to pinpoint tail-latency degradations before users are impacted.

● Status Code Distribution

Categorical bar charts contrasting successful 2xx traffic against client rate-limiting (429), security blocks (403), and upstream 5xx outages.

● Traffic Volume & RPS Trends

Monitors requests-per-second (RPS) velocity to identify organic traffic spikes, scheduled batch jobs, or distributed denial-of-service (DDoS) bursts.

● Monochrome Tooltips & Clean SVGs

Built with Recharts using high-contrast minimalist charts that render seamlessly in both dark room operations centers and light office mode.

MODULE 11 • SECURITY & FIREWALL

CIDR IP Policies & Denylisting

The IP Policies module enforces layer-3/layer-4 access boundaries at the gateway edge. It supports single IP addresses, CIDR network masks (e.g. `10.0.0.0/8` , `192.168.1.0/24`), permanent bans, and temporary bans with auto-expiration.

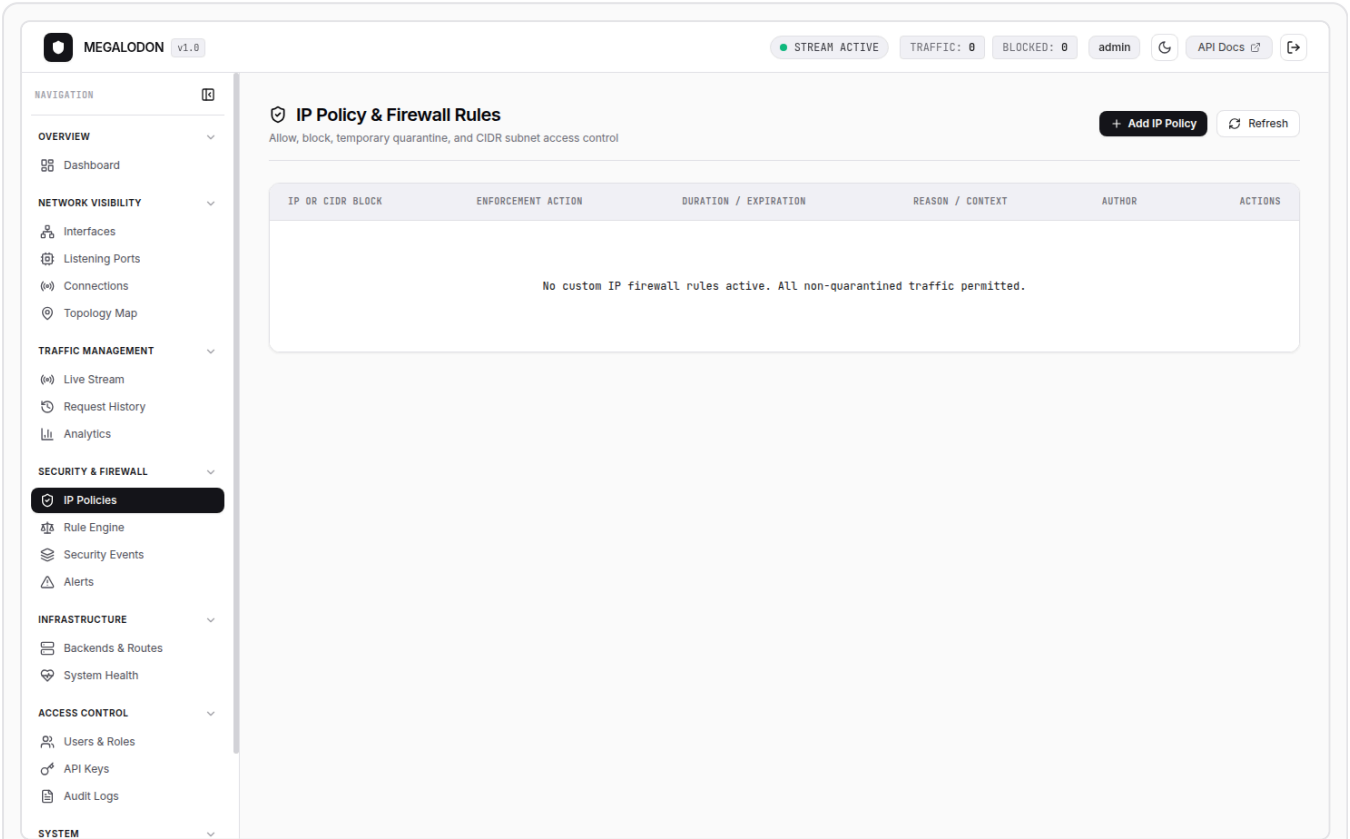


Figure 11.1: CIDR IP Access Control & Firewall Policy Table

Route: /security/ip-policies • In-Memory & Redis Cache

● Permanent & Temporary Blocks

Supports permanent mitigation or time-bounded quarantine (e.g. 15-minute cool-off for brute-force scanners) with automatic release.

● Subnet CIDR Range Support

Evaluates incoming client addresses against complete CIDR subnet masks using optimized bitwise network masks in sub-microsecond time.

● Allowlist (Whitelist) Overrides

Explicit `ALLOW` policies can exempt trusted partner APIs, health-check probes, or office IPs from general rate limits.

● Redis Dynamic Synchronization

Policies created or revoked in the UI sync instantly to Redis and are hot-reloaded into gateway memory with zero restart required.

MODULE 12 • SECURITY & FIREWALL

Visual Security Rule Engine

A declarative visual policy builder allowing security administrators to construct complex multi-condition boolean rules without writing code. Rules evaluate incoming HTTP methods, paths, headers, query strings, and body signatures.

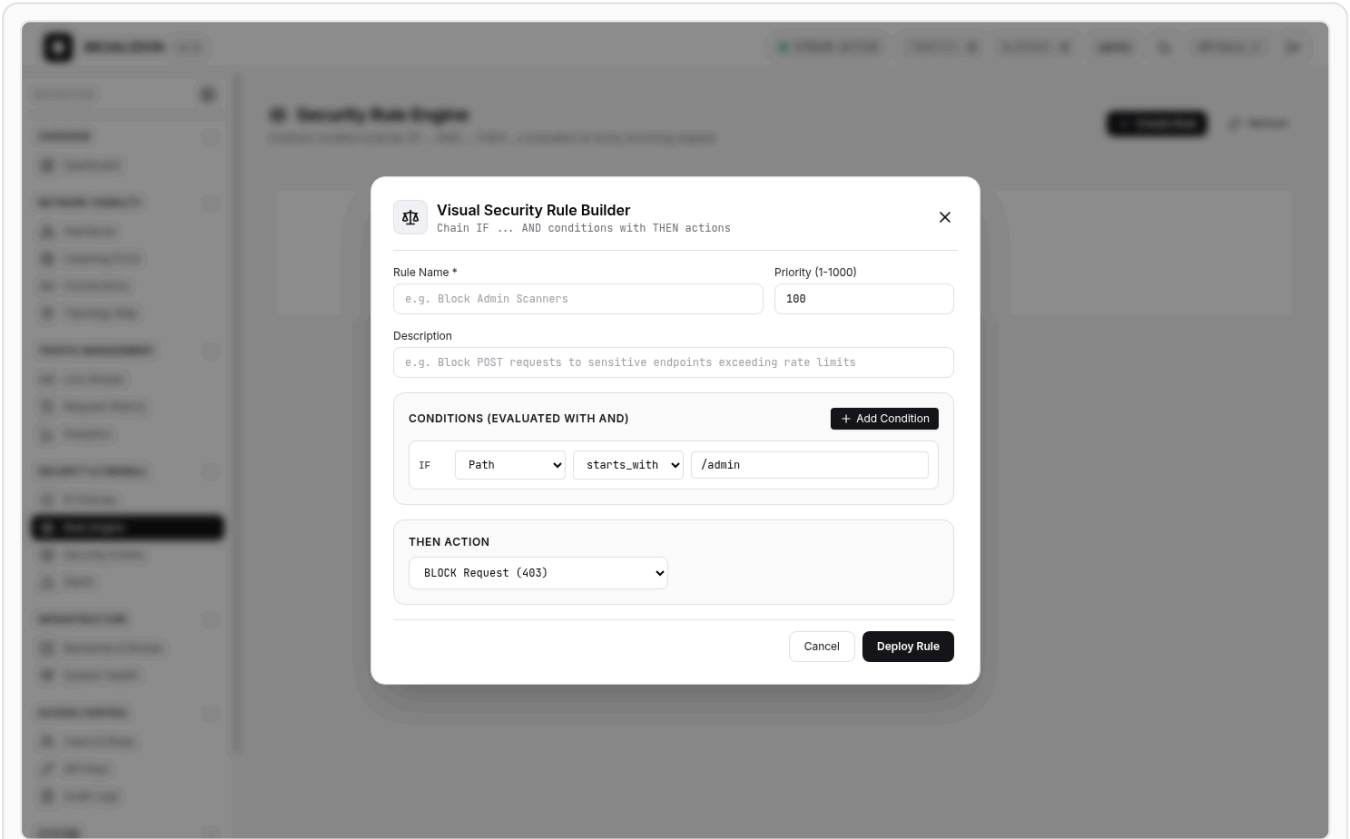


Figure 12.1: Visual Security Rule Builder Modal (Portaled document.body)

Route: /security/rules • Visual Rule Builder

• Boolean Condition Chaining

Chain multiple conditions evaluated with logical `AND` across fields: `Path`, `Method`, `Header`, `Query`, and `Client IP`.

• Flexible Operators

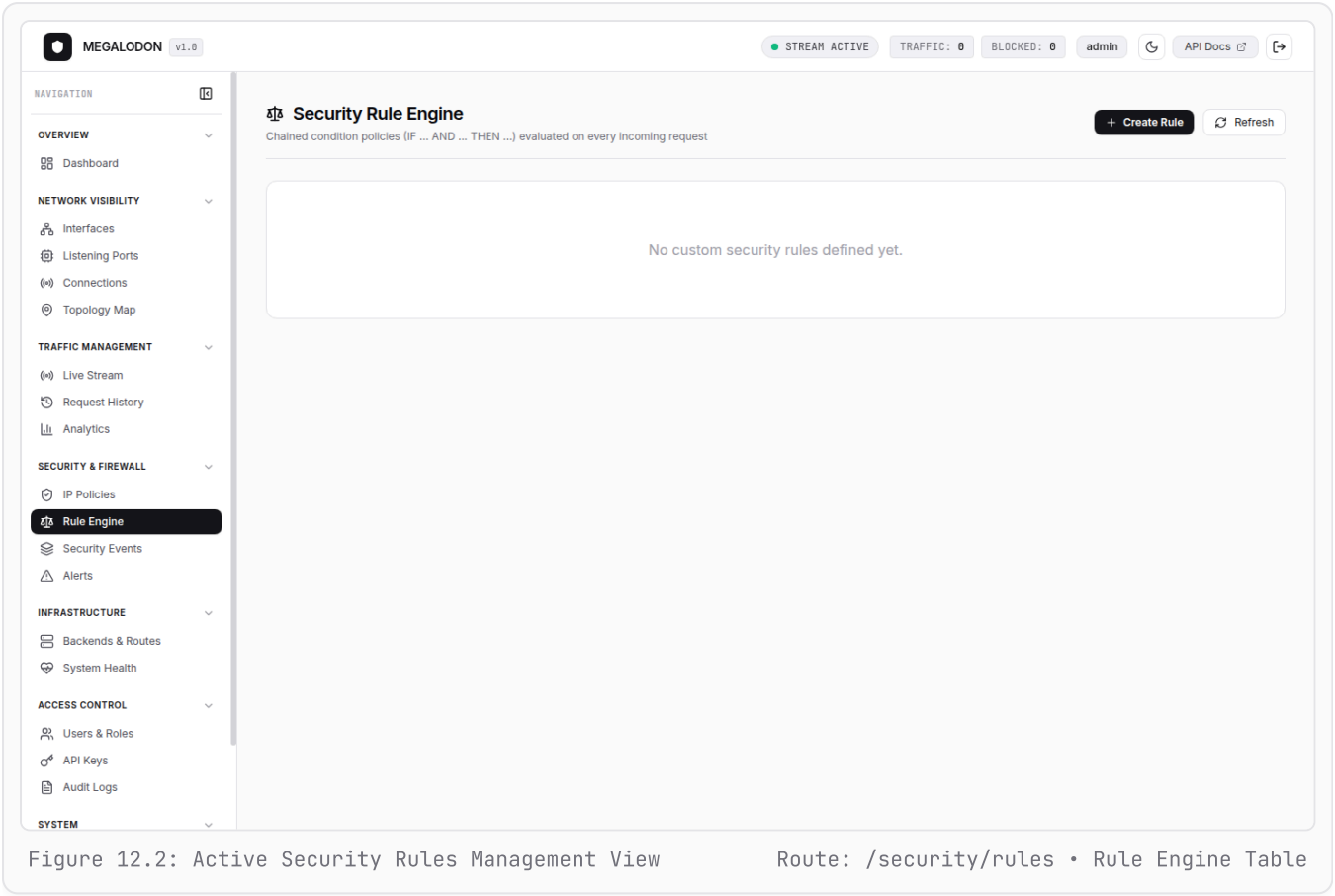
Operators include `equals`, `contains`, `starts_with`, `ends_with`, `regex_match`, and `in_cidr`.

• Granular THEN Actions

Actions include: `BLOCK Request (403)`, `RATE_LIMIT (429)`, `CUSTOM_STATUS (Custom Code)`, or `LOG_ALERT`.

• Priority-Based Execution

Configurable numeric priority (1 - 1000). Rules are evaluated sequentially; the highest-priority matching rule terminates evaluation.



MODULE 13 • SECURITY & FIREWALL

Security Threat Events & Signatures

When suspicious traffic or malicious patterns are intercepted by Megalodon workers, structured security event records are created. This feed acts as the primary intrusion detection system (IDS) audit log.

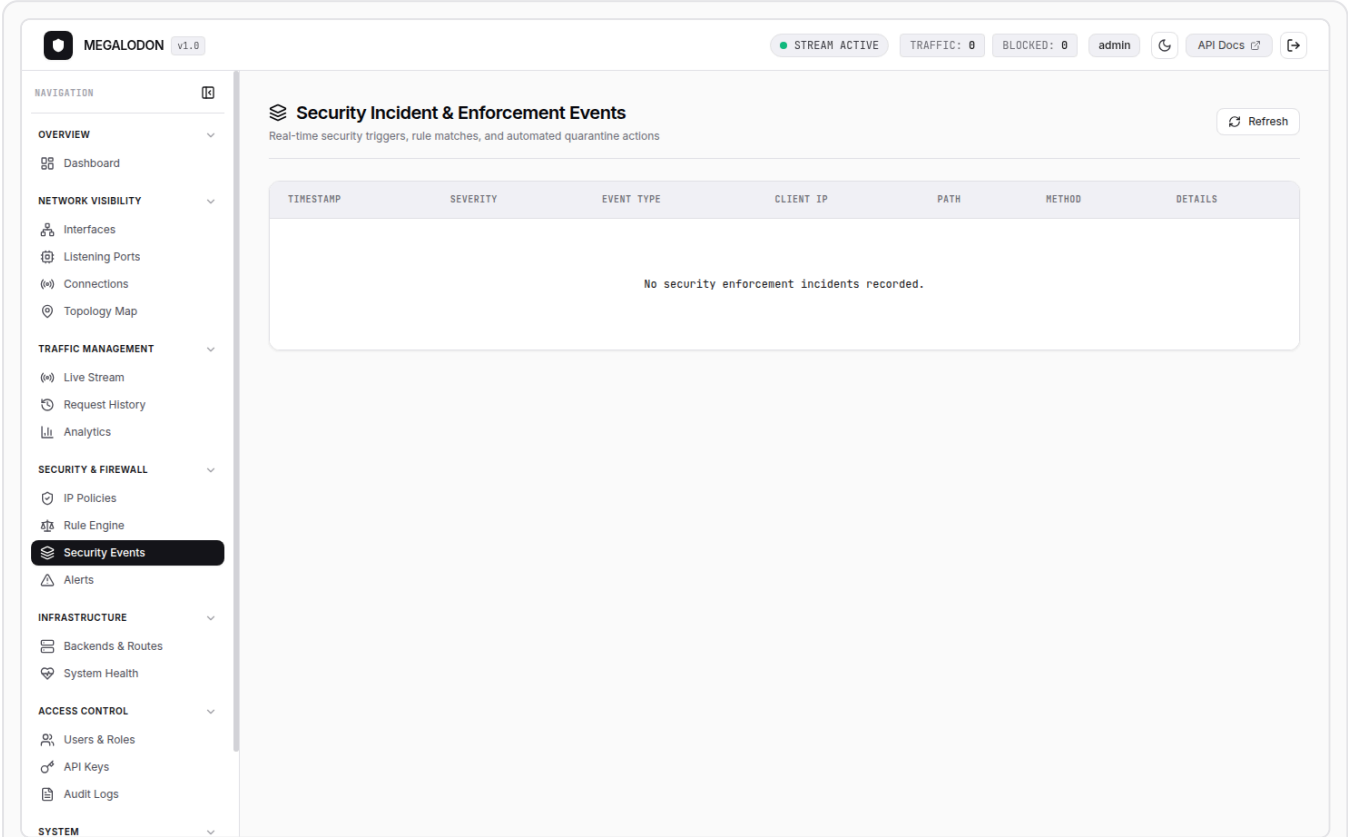


Figure 13.1: Intercepted Security Threat Events & Threat Classification

Route: /security/events • Intrusion Detection Log

● Automated Signature Detection

Detects common web attack vectors: SQL Injection (SQLi), Cross-Site Scripting (XSS), Directory Traversal (.. / .. /), and command injection attempts.

● Severity Level Scoring

Events are categorized with strict severity levels: **CRITICAL**, **HIGH**, **MEDIUM**, and **LOW** for rapid triage.

● Attack Payload Forensic Capture

Extracts offending URI substrings, malicious headers, and raw user-agent strings for security incident documentation and evidence collection.

● One-Click Mitigation

Quick action controls allow analysts to directly denylist the attacker's IP or generate an automated visual security rule from the event details.

Real-Time Security Incident Alerts

The Alerts console aggregates actionable, high-priority notifications requiring immediate operator intervention. Generated by the `megalodon-alert-worker` when attack frequencies exceed threshold baselines.

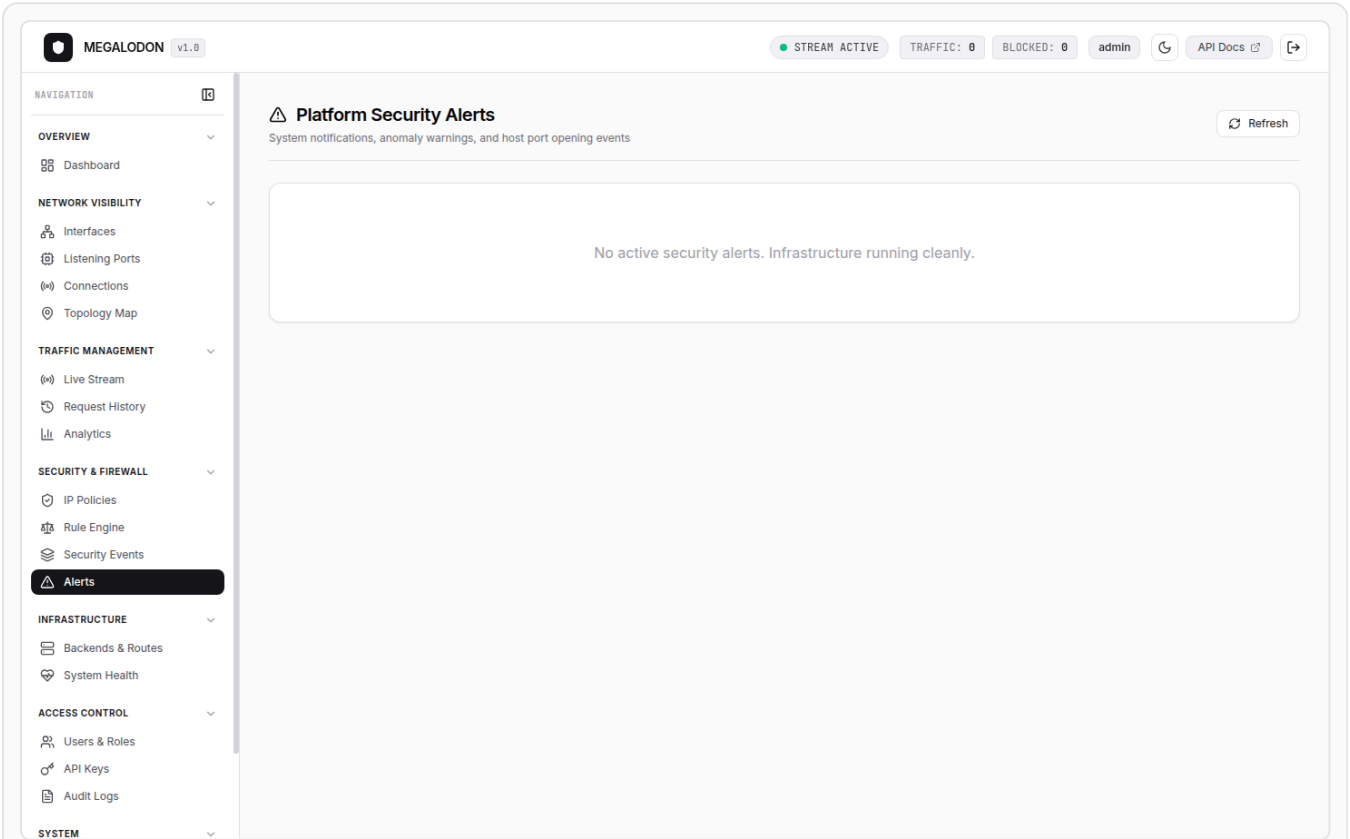


Figure 14.1: Real-Time Security Incident & Operator Alert Queue Route: /security/alerts • Alert Dispatcher

● Incident Acknowledgment & Resolution

Operators can transition alert lifecycles from `OPEN` to `ACKNOWLEDGED` to `RESOLVED` with full timestamped audit tracking.

● Volumetric Threshold Alarms

Fires automated alerts when specific endpoints experience abnormal 4xx spikes or when an IP triggers repeated rule rejections.

● Incident Severity Filters

Quickly toggle between critical system-level alarms and informational warnings to prevent alert fatigue during operational shifts.

● Clean Empty-State Handling

When all incidents are resolved, Megalodon displays a minimalist verified "All systems clear" reassurance card.

Upstream Backends & Reverse Proxy Routing

Configures the reverse proxy routing topology of the Megalodon Ingress Engine. Maps external incoming URL paths to internal container ports or microservice server clusters.

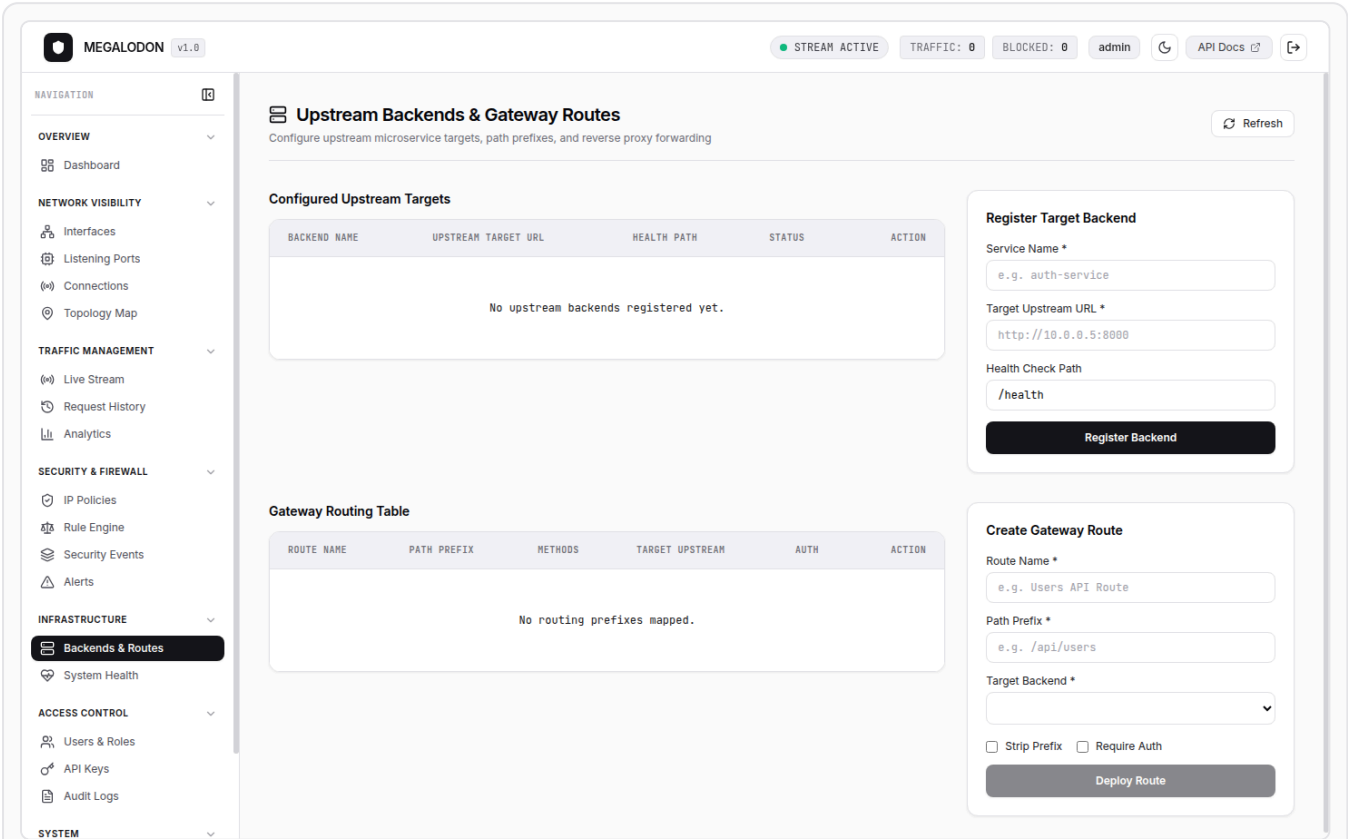


Figure 15.1: Upstream Backends Pool & Reverse Proxy Routing Table

● Backend Pool Management

Register upstream targets (e.g. `http://demo-users:5001`), internal auth services, payment APIs) with target host, port, and health check intervals.

● Upstream Health Probes

Continuously executes active HTTP heartbeat probes against upstream targets. Unhealthy nodes are seamlessly taken out of proxy rotation.

● Path Prefix Matching

Routes requests based on URL prefix: e.g. requests matching `/api/v1/users/*` automatically route to the demo-users upstream service.

● Hot Route Reloading

Routes and backends created through the UI take effect immediately in the Gateway without server restarts or dropped connections.

MODULE 16 • INFRASTRUCTURE

System Health & Infrastructure Dependencies

Monitors the real-time operational status of all core backing services that Megalodon relies upon. Detects service degradation, connection drops, or latency bottlenecks across the stack.

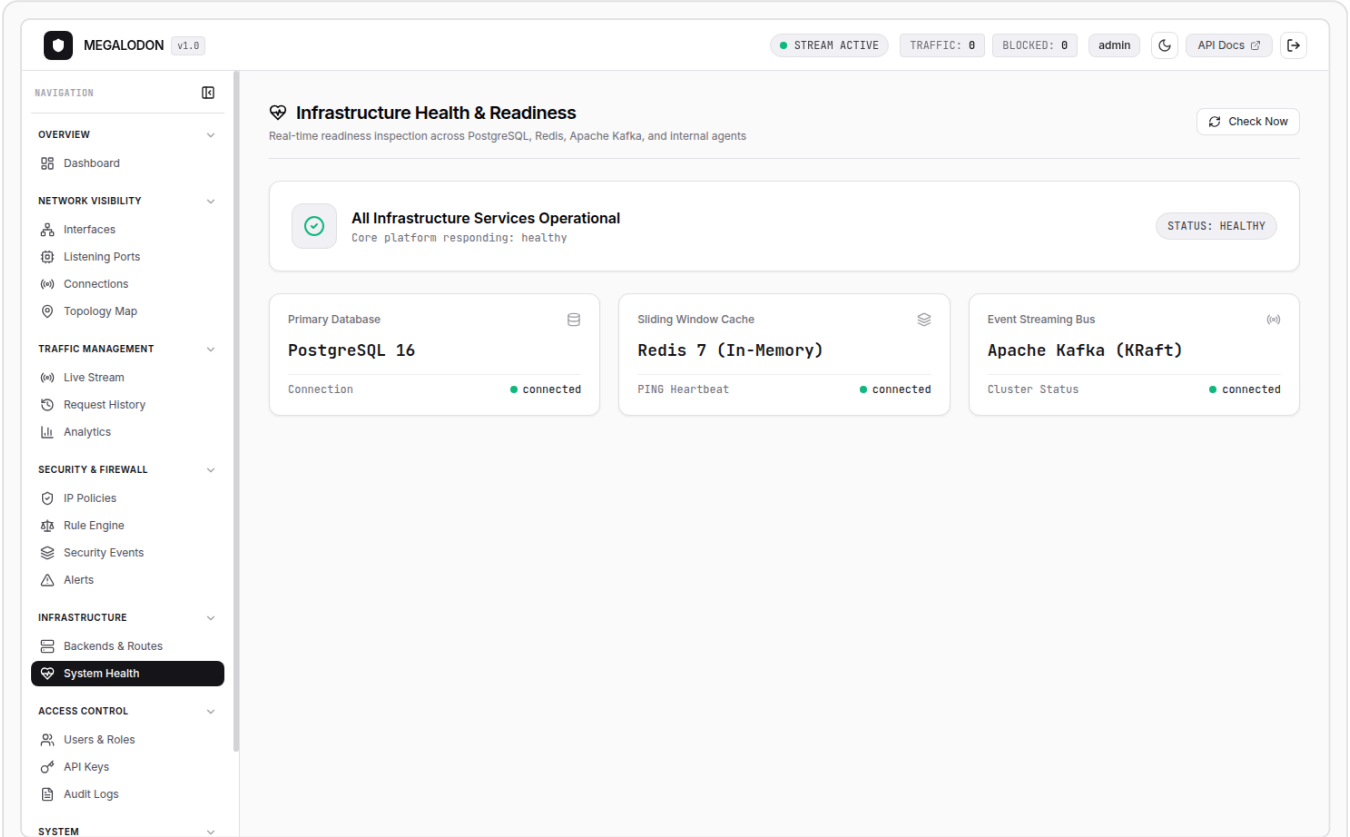


Figure 16.1: Distributed Infrastructure Health & Dependency Status Cards Route: /infrastructure/health • Health Probes

● PostgreSQL Database

Monitors relational database connection pool health, query response latency, and pending migration states.

● Redis In-Memory Cache

Tracks Redis memory utilization, ping-pong heartbeat responsiveness, and rate-limiting key eviction rates.

● Apache Kafka Event Broker

Verifies broker connectivity, topic partition readiness, and consumer group processing lag for traffic streams.

● Host Network Agent Daemon

Ensures the background host telemetry daemon is actively reporting kernel socket updates and interface changes.

MODULE 17 • ACCESS CONTROL

Operator Accounts & Role-Based Access (RBAC)

Implements least-privilege security administration. Administrators can provision operator accounts with scoped roles, enforce password policies, and instantly disable compromised credentials.

MEGALODON v1.0

STREAM ACTIVE TRAFFIC: 0 BLOCKED: 0 admin API Docs

NAVIGATION

- OVERVIEW
 - Dashboard
- NETWORK VISIBILITY
 - Interfaces
 - Listening Ports
 - Connections
 - Topology Map
- TRAFFIC MANAGEMENT
 - Live Stream
 - Request History
 - Analytics
- SECURITY & FIREWALL
 - IP Policies
 - Rule Engine
 - Security Events
 - Alerts
- INFRASTRUCTURE
 - Backends & Routes
 - System Health
- ACCESS CONTROL
 - Users & Roles**
 - API Keys
 - Audit Logs
- SYSTEM

Role-Based Access Control (RBAC)

Manage operator accounts, administrative permissions, and auditor access

Active Operator Accounts

USERNAME	EMAIL ADDRESS	ASSIGNED ROLES	STATUS	ACTION
admin	admin@megalodon.local	ADMIN	ACTIVE	

Create Operator Account

Username *

alice

0/3+

Email *

alice@megalodon.local

Password *

.....

SECURITY POLICY CHECKLIST

✗ Username 3+ chars

✗ Password 8+ chars

✗ Upper & lowercase

✗ Number & symbol

Role Permission

VIEWER (Read Only)

Create Account

Figure 17.1: Operator Account Management & RBAC Roles

Route: /access/users • User Administration

● Role: ADMIN

Full administrative permissions: user provisioning, rule deployment, policy modifications, backend management, and system settings.

● Role: ANALYST

Read-only visibility across dashboards, live traffic streams, network maps, and analytics with rule inspection rights.

● Role: OPERATOR

Operational rights to acknowledge security alerts, ban offending IPs, and pause live traffic streams without system config privileges.

● Bcrypt Cryptographic Hashing

All operator passwords are encrypted using multi-round salted Bcrypt hashes. Raw credentials are never stored or logged.

Megalodon API Security Platform — Administrator Manual

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MODULE 18 • ACCESS CONTROL

Machine-to-Machine API Keys & Scopes

Allows automated services, CI/CD deployment pipelines, and external integrations to authenticate programmatically against the Megalodon API without human interactive credentials.

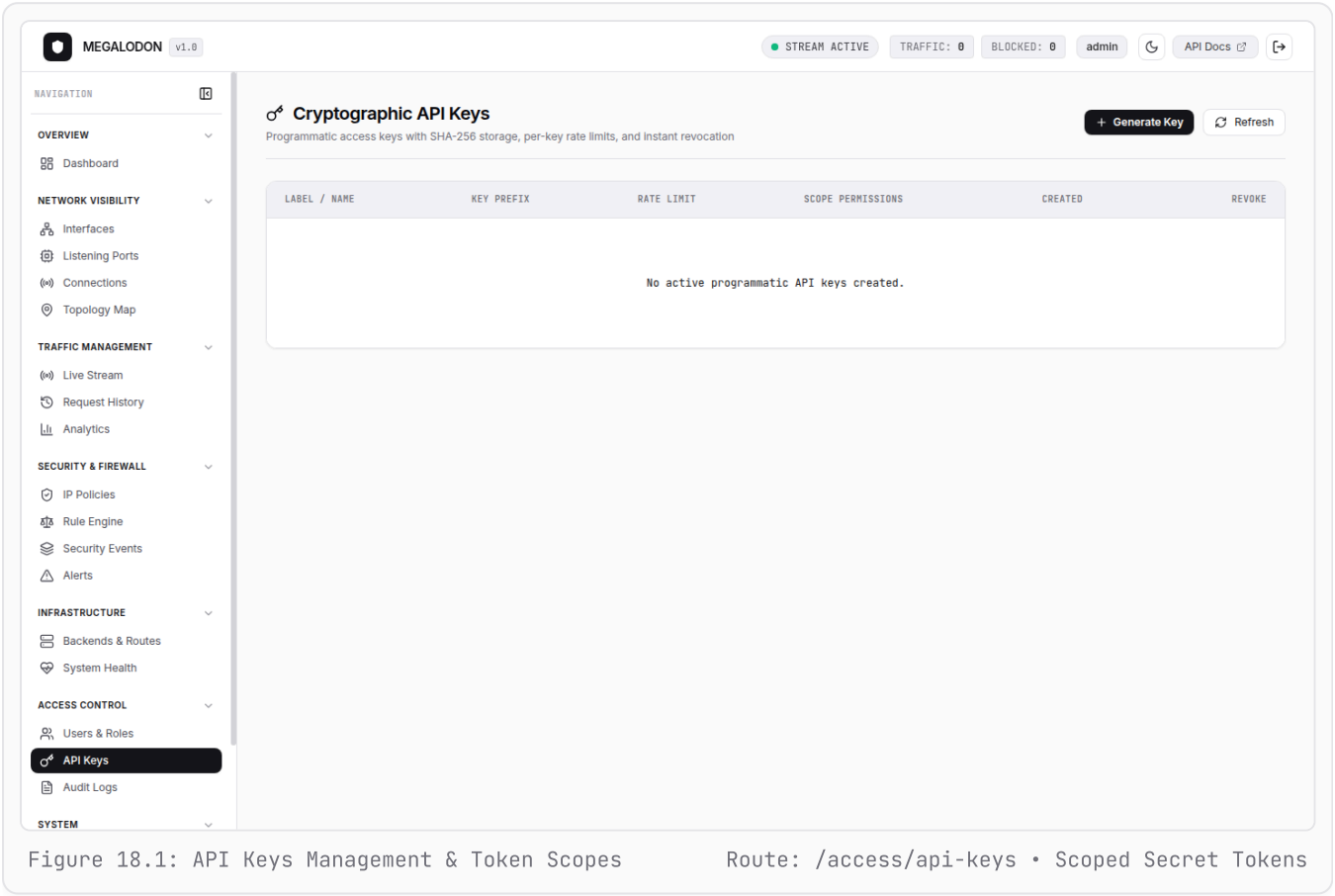


Figure 18.1: API Keys Management & Token Scopes

Route: /access/api-keys • Scoped Secret Tokens

● Cryptographic Secret Generation

Generates 256-bit cryptographically secure API tokens with clear prefix identification (e.g. `sen_live_...`). Secret is shown only once at creation.

● Granular Token Scopes

Restrict token capabilities using granular scopes: `read:metrics`, `write:policies`, `admin:full` to prevent privilege escalation.

● Per-Key Rate Limiting

Attach individual requests-per-minute (RPM) limits to specific tokens to prevent rogue automated scripts from overwhelming the gateway.

● Instant Token Revocation

One-click revocation immediately invalidates the token across all gateway instances, terminating active programmatic sessions.

MODULE 19 • ACCESS CONTROL

Tamper-Evident Security Audit Logs

Maintains a strictly immutable audit record of every administrative action executed on the platform. Satisfies SOC2, ISO27001, and HIPAA compliance requirements for traceability.

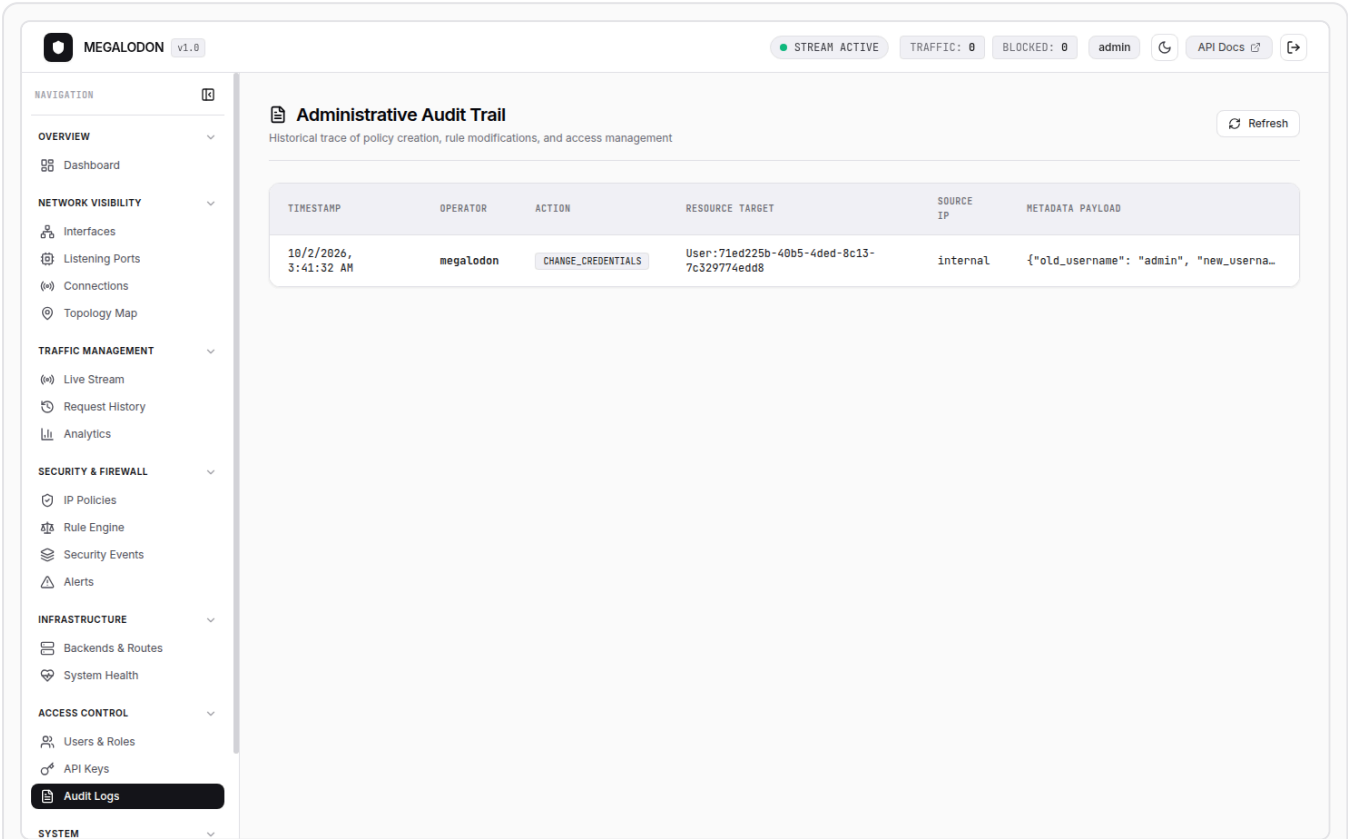


Figure 19.1: Cryptographic Administrative Action Audit Route: /access/audit-logs • Immutable Audit Trail

● Action Traceability

Logs exact action types: `RULE_CREATED`, `IP_BLOCKED`, `USER_MODIFIED`, `ROUTE_DELETED`, and `CONFIG_CHANGED`.

● Actor & IP Attribution

Records the initiating operator username, client source IP address, user-agent, and exact UTC millisecond timestamp.

● Structured JSON Metadata

Stores before-and-after change diffs in structured JSON payloads, allowing auditors to inspect previous and new configuration values.

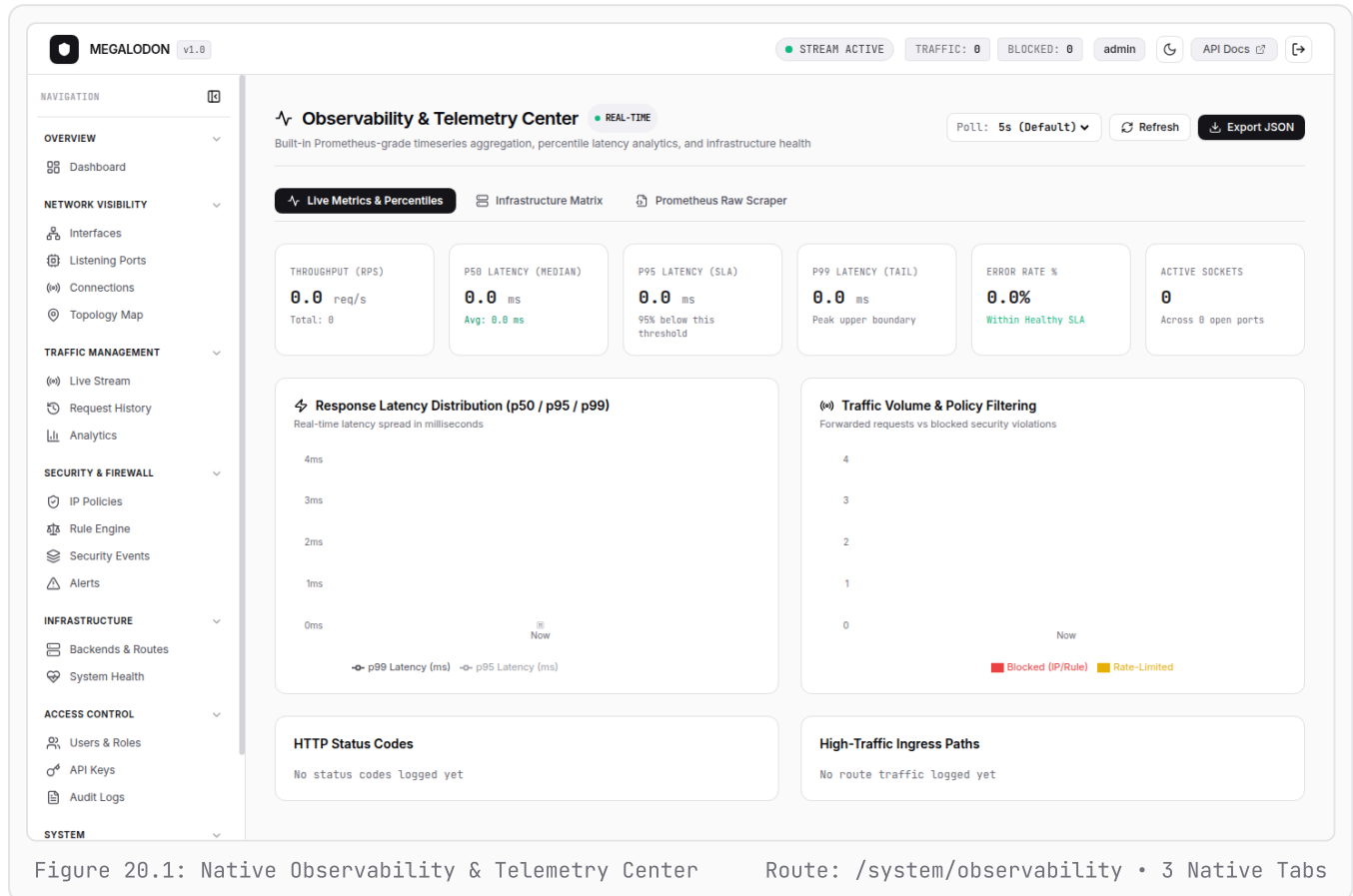
● Immutable Storage

Audit log entries cannot be modified or deleted via the UI or standard API endpoints, ensuring non-repudiation of security events.

MODULE 20 • SYSTEM OBSERVABILITY

Built-in Observability Center & Real-Time Telemetry

Megalodon features a native, zero-overhead Observability Center embedded directly into the Web Panel—eliminating the need for separate Prometheus and Grafana containers. It provides full-fidelity timeseries analytics, latency distribution curves, infrastructure health checks, and OpenMetrics scrape inspection.



● P50 / P95 / P99 Latency Percentiles

Computes dynamic statistical percentiles (P50, P95, P99) across sliding-window time buckets with interactive Recharts AreaCharts and SLA health status indicators.

● Traffic Breakdown & Status Meters

Categorizes live ingress traffic into Allowed, Blocked, and Rate-Limited volume alongside real-time HTTP status distribution meters (2xx, 3xx, 4xx, 5xx).

● Infrastructure Health Matrix

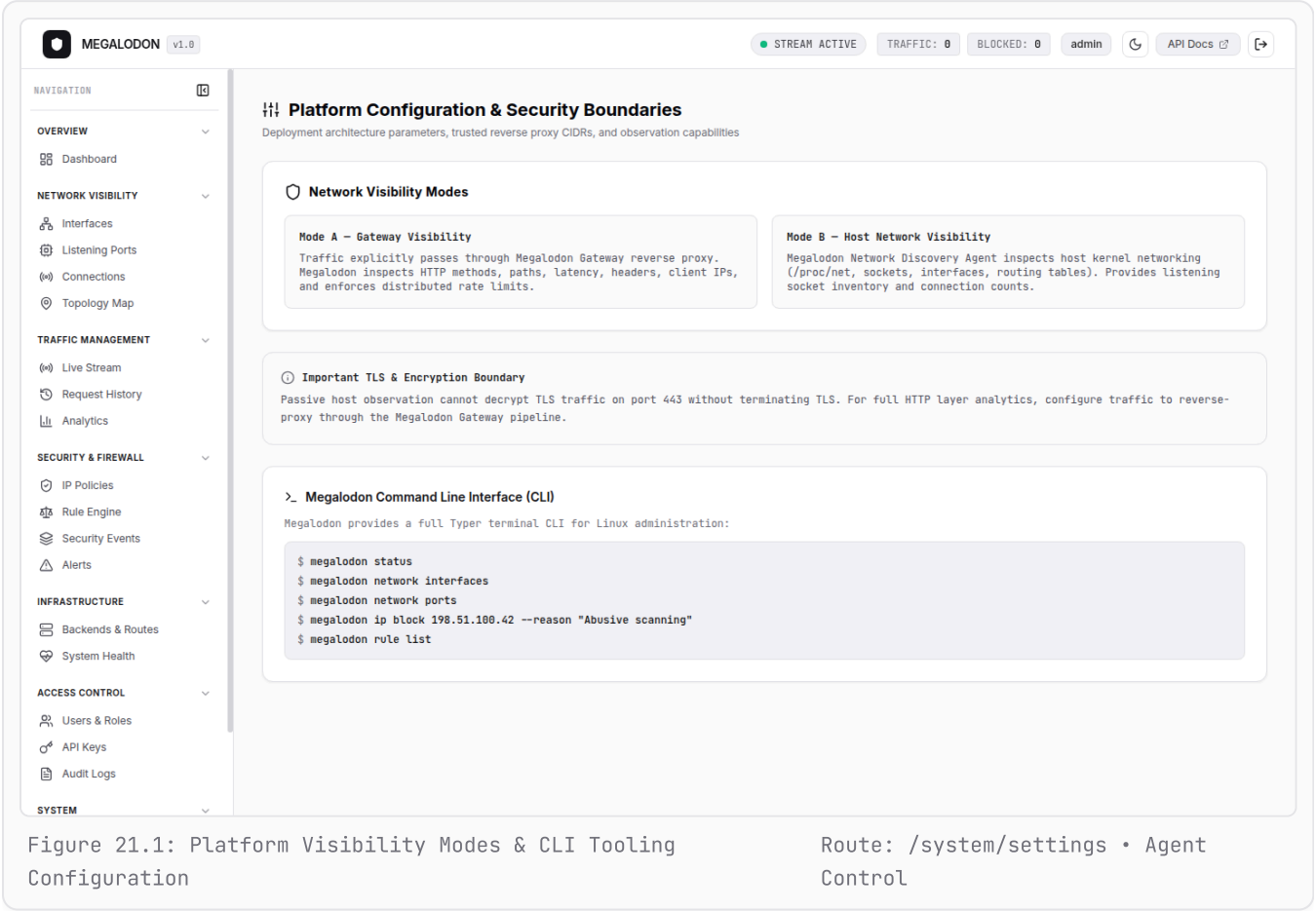
Monitors operational readiness, ping latency, CPU %, memory consumption (MB and %), and system uptime across Core API, PostgreSQL, Redis, and Kafka.

● Prometheus Raw Scraper

In-browser terminal viewing raw OpenMetrics telemetry from `/api/v1/metrics/text` with real-time substring filtering, one-click copy, and JSON export.

System Settings & Network Visibility Modes

Configures host collection depth, kernel packet capture modes, and provides command-line tool snippets for interacting with the Megalodon CLI & Tooling.



● **Visibility Mode: STANDARD**

Lightweight kernel Netlink socket auditing with low CPU overhead (< 0.5%). Recommended for baseline production deployments.

● **Visibility Mode: PROMISCUOUS**

Deep packet capture mode that inspects packet headers across non-local broadcast interfaces for network-wide intruder detection.

● **Visibility Mode: FORENSIC**

Comprehensive logging mode recording raw payload fragments and full TCP handshake metadata during active incident response.

● **Megalodon CLI & Tooling**

Native terminal utility enabling terminal-based socket sweeps and interface diagnostics directly on Linux host servers.

TERMINAL QUICK COMMANDS REFERENCE

```
# Check Megalodon platform health and latency  
megalodon health
```

```
# Run manual socket scan across all host interfaces  
megalodon network ports
```

```
# Verify Redis policy cache synchronization  
docker exec -it megalodon-redis redis-cli ping
```

```
# Inspect live Kafka ingress stream topics  
docker exec -it megalodon-kafka /opt/kafka/bin/kafka-topics.sh --bootstrap-server localhost:9092
```

MODULE 22 • APPENDIX

REST API Reference & Operational Cheatsheet

Megalodon features a comprehensive RESTful API exposed at `http://localhost:8000/api/v1` (with public Swagger UI disabled in production for security hardening).

METHOD	ENDPOINT URI	DESCRIPTION & PARAMETERS
POST	<code>/api/v1/auth/login</code>	Authenticate operator credentials, returns JWT bearer access token
GET	<code>/api/v1/network/interfaces</code>	Fetch discovered host network interfaces, MACs, link states & MTUs
GET	<code>/api/v1/network/ports</code>	List active listening TCP/UDP sockets, bound processes & connections
GET	<code>/api/v1/network/connections</code>	List active established TCP connections with remote client addresses
GET	<code>/api/v1/network/map</code>	Get 3-stage ingress topology graph nodes and connection links
GET	<code>/api/v1/traffic/history</code>	Query historical request ingress logs with status code filters
GET	<code>/api/v1/traffic/analytics</code>	Retrieve P50, P95, P99 latency percentiles & status code distributions
GET/POST	<code>/api/v1/security/rules</code>	List, create, update, or toggle custom visual security rules
GET/POST	<code>/api/v1/security/ip-policies</code>	Manage CIDR IP denylists, allowlists, and temporary blocks
GET	<code>/api/v1/security/events</code>	Query detected threat events, attack signatures, and severities
GET/POST	<code>/api/v1/security/alerts</code>	Inspect active security alerts, transition incident status
GET/POST	<code>/api/v1/infrastructure/backends</code>	Register upstream backend pools and reverse proxy routing paths
GET	<code>/api/v1/infrastructure/health</code>	Deep health check of PostgreSQL, Redis, Kafka, and Network Agent
GET/POST	<code>/api/v1/access/users</code>	User management and RBAC role assignment
GET/POST	<code>/api/v1/access/api-keys</code>	Generate and revoke machine-to-machine API access tokens
GET	<code>/api/v1/access/audit-logs</code>	Immutable administrative action trail with before/after diffs
WS	<code>/api/v1/ws/traffic</code>	WebSocket streaming endpoint for real-time gateway ingress events

END OF MANUAL & COURSE COMPLETION

Congratulations! You have completed the comprehensive Megalodon API Security Platform training manual. All modules, interfaces, and architecture layers are now operational and ready for enterprise production traffic.